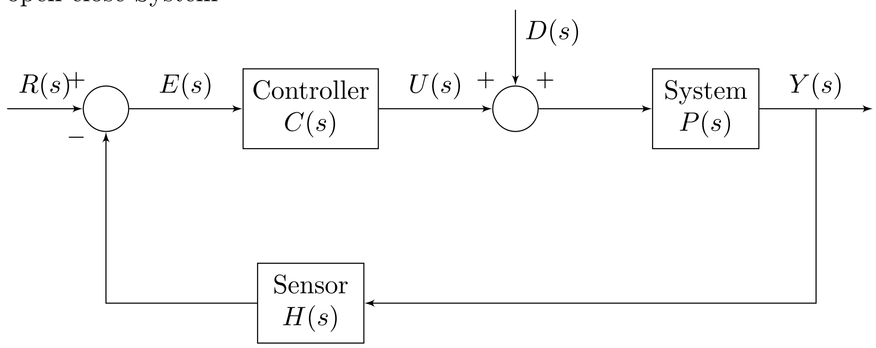
System
Disturbance
Feedback loop
open close System



1 Linear Constant Coefficient Ordinary Differential Equations (LCC-ODE)

A differential equation of the form

$$\sum_{k=0}^{n} a_k \frac{d^k y(t)}{dt^k} = \sum_{m=0}^{M} b_m \frac{d^m r(t)}{dt^m},$$

where $y(t)$ is the output (response) of the system and $r(t)$ is the input to the system; and a_i and b_i are constant.

2 Laplace Transform

The Laplace transform of a function $f(t)$, defined for all real numbers $t \geq 0$, is the function $F(s)$, which is a unilateral transform defined by

$$\mathcal{L}[f](s) = F(s) = \int_{0^-}^{\infty} f(t)e^{-st}dt,$$

where $s = \sigma + \imath\omega$ complex frequency-domain parameter, and, 0^- is the shorthand notation for

$$\lim_{\varepsilon \to 0^+} \int_{-\varepsilon}^{\infty}.$$

2.1 Region of Convergence (RoC)

2.2 Properties of Laplace Transform

2.3 Standard Laplace Transform Pairs

Function	Time Domain $f(t)$	Laplace Domain $F(s)$	ROC
Impulse	$\delta(t)$	1	All s
Step	$u(t)$	$\frac{1}{s}$	$\Re(s) > 0$
Ramp	$tu(t)$	$\frac{1}{s^2}$	$\Re(s) > 0$
Power	$t^n u(t)$	$\frac{n!}{s^{n+1}}$	$\Re(s) > 0$
Exponential	$e^{-at}u(t)$	$\frac{1}{s+a}$	$\Re(s) > -a$
Sine	$\sin(\omega t)u(t)$	$\frac{\omega}{s^2+\omega^2}$	$\Re(s) > 0$
Cosine	$\cos(\omega t)u(t)$	$\frac{s}{s^2+\omega^2}$	$\Re(s) > 0$

3 Second-Order System Analysis

Consider a linear time-invariant (LTI) second-order system described by the transfer function $G(s)$:

$$G(s) = \frac{b}{s^2 + as + b}$$

where a and b are constant coefficients. To analyze the dynamic behavior, we map these coefficients to standard physical parameters.

Definition 3.1 (Natural Frequency, ω_n): *The natural frequency, ω_n, represents the frequency at which the system would oscillate in the absence of damping. It is defined as:*

$$\omega_n := \sqrt{b}$$

Definition 3.2 (Damping Ratio, ζ): *The damping ratio, ζ, is a dimensionless parameter that characterizes the rate at which oscillations decay. It is related to the system coefficients by:*

$$\zeta := \frac{a}{2\omega_n} = \frac{a}{2\sqrt{b}}$$

Substituting these parameters back into the original expression yields the **canonical form** of the second-order transfer function:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

The poles of the system, denoted as $s_{1,2}$, are the roots of the characteristic equation $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$. Using the quadratic formula, the poles are given by:

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

4 Underdampled Second Order System

$$Y(s) = \frac{1}{s}\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2} = \frac{A_1}{s} + \frac{A_2 s + A_3}{s^2+2\zeta\omega_n s+\omega_n^2}$$

$$Y(s) = \frac{1}{s} - \frac{s+\zeta\omega_n}{(s+\zeta\omega_n)^2+\omega_n^2(1-\zeta^2)} - \frac{(\zeta/\sqrt{1-\zeta^2})\omega_n\sqrt{1-\zeta^2}}{(s+\zeta\omega_n)^2+\omega_n^2(1-\zeta^2)}$$

$$\omega_d \coloneqq \omega_n\sqrt{1-\zeta^2}$$

$$y(t) = \left(1 - e^{-\zeta\omega_n t}\cos(\omega_d t) - \frac{\zeta}{\sqrt{1-\zeta^2}}e^{-\zeta\omega_n t}\sin(\omega_d t)\right)u(t)$$

Definition 4.1 (Rise time, T_r): *The time required for the waveform to go from* 0.1 *of the final value to* 0.9 *of the final value.*

$$T_r = \frac{\pi-\theta}{\omega_n\sqrt{1-\zeta^2}}, \quad \theta = \arctan\left(\frac{\sqrt{1-\zeta^2}}{\zeta}\right)$$

Definition 4.2 (Peak time, T_p): *The time required to reach the first, or maximum, peak.*

$$T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$$

Definition 4.3 (Max overshoot, OS): *The amount that the waveform overshoots the steady-state, or final, value at the peak time, expressed as a ratio of the steady-state value.*

$$OS = \exp\left(-\zeta\pi/\sqrt{1-\zeta^2}\right)$$

Definition 4.4 (Settling time, T_s): *The time required for the transient's damped oscillations to reach and stay within* $\pm 2\%$ *of the steady-state value.*

$$T_s = \frac{-\ln\left(0.02\sqrt{1-\zeta^2}\right)}{\zeta\omega_n} \approx \frac{4}{\zeta\omega_n}$$

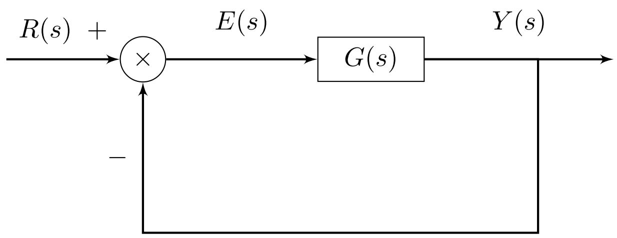
5 Second Order System With Additional Pole

6 Feedback Loop

Definition 6.1 (Feedback Loop Transfer Function):

$$T(s) \coloneqq \frac{Y(s)}{R(s)} = \frac{G(s)}{1+G(s)H(S)}$$

7 Unity Feedback Steady State Error



Consider a unity feedback system with forward transfer function $G(s)$. The closed-loop structure is

$$Y(s) = G(s)E(s), \qquad E(s) = R(s) - Y(s).$$

Definition 7.1 (Steady State Error): *Suppose the closed-loop system is asymptotically stable and the limits exist. The steady-state error is defined as*

$$e_{ss} := \lim_{t\to\infty} e(t).$$

If $e(t)$ admits a Laplace transform $E(s)$ and the Final Value Theorem applies, then

$$e_{ss} = \lim_{s\to 0} sE(s).$$

Theorem 7.2 (Steady-State Error for Unity Feedback): *For a unity feedback system with forward transfer function $G(s)$, if the closed-loop system is asymptotically stable and the Final Value Theorem applies, then*

$$e_{ss} = \lim_{s\to 0} \frac{sR(s)}{1+G(s)}.$$

Proof. From the unity feedback structure,

$$E(s) = R(s) - Y(s).$$

Since $Y(s) = G(s)E(s)$, we substitute to obtain

$$E(s) = R(s) - G(s)E(s).$$

Rearranging,

$$E(s)(1 + G(s)) = R(s).$$

Hence,

$$E(s) = \frac{R(s)}{1+G(s)}.$$

Assume:

1. The closed-loop system is asymptotically stable.
2. $sE(s)$ has no poles in the closed right half-plane except possibly at $s = 0$.

Then the Final Value Theorem applies, and

$$e_{ss} = \lim_{t \to \infty} e(t) = \lim_{s \to 0} sE(s).$$

Substituting the expression for $E(s)$,

$$e_{ss} = \lim_{s \to 0} s \frac{R(s)}{1 + G(s)}.$$

This proves the result. $\square$

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